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Peer-graded Assignment: Week 2: Multiple Linear Regression

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multilinear regression

 by Anonymous Learner

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PROMPT

In **question 1d** in your Module 2 Jupyter notebook lab, you trained models with polynomial degrees up to 20, printing out each polynomial degree and r-squared value.

Inspect the model summary from each model and discuss your findings. You filled in the highest order model that makes sense in `sound_degree`. Why don't higher-order models make sense?

Type your answer in the rich text field.

1. **Increasing Complexity:** As you increase the polynomial degree, the complexity of the model increases. Higher-degree polynomials can capture complex relationships in the data, but they can also lead to overfitting, where the model fits the noise in the data rather than the underlying pattern.
2. **Overfitting:** When you inspect the model summaries, you may notice that as the polynomial degree increases, the model starts fitting the training data very closely, resulting in high R-squared values. However, this doesn't necessarily mean the model will perform well on unseen data. Overfit models are highly sensitive to noise and may not generalize well.

